Lya Chiñas Serrano

PhD Student, Machine Learning for Biology

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Website
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she/her

Education

'24–Now **PhD in Chemical Engineering (direct entry)**, *University of Toronto*, Toronto, Canada
Laboratory for Metabolic Systems Engineering. Advisor: *Prof. Radhakrishnan Mahadevan.*
Coursework: Applied Deep Learning (ECE1508); Deep Learning: Theory & Data Science (MAT1510); Creative Applications of NLP (ECE1786); Data Science and Analytics (MIE1624).

'20–'24 **BSc in Biotechnology Engineering**, *Monterrey Institute of Technology and Higher Education (ITESM)*, Puebla, Mexico
GPA 4.0. First in Class Honors.

Research Focus

Topics: Protein and genome language models, enzyme function and kinetics prediction, bacterial phenotype prediction, genome-scale metabolic models.

Research Experience

'24–Now **Graduate Student Researcher**, *Laboratory for Metabolic Systems Engineering, University of Toronto*
Lead machine learning research that predicts bacterial phenotypes, from data collection to model evaluation.
- Predicting enzyme kinetics and bacterial growth from protein and genome language model embeddings (ESM, Evo2).
- Building benchmarks and evaluation splits (sequence similarity, held-out clades, unseen conditions).
- Curating genome, proteome, phenotype, and culture media data from public databases.
- Turning high-throughput experimental data (plate-reader growth curse) into curated datasets.
- Automating literature research and data curation with LLM agents (LangGraph).

'23 **Mitacs Globalink Research Intern**, *Laboratory for Metabolic Systems Engineering, University of Toronto*
Project on kinetic parameter-constrained genome-scale metabolic models. Built a Python framework that refines these models, using constraint-based modeling and linear programming.

Publications

'26 R. A. Barghout, **L. C. Serrano**, B. Sanchez-Lengeling, R. Mahadevan. kinGEMs: A Robust and Scalable Framework for Resource-Constraint Models through Stochastic Tuning of Deep Learning-Predicted Kinetic Parameters. *PLOS Computational Biology* (in press). doi:10.64898/2026.03.14.711833

'26 R. A. Barghout, M. Kirby, A. Zheng, **L. Chiñas**, M. Mohammadi, Z. Xu, B. Sanchez-Lengeling, R. Mahadevan. Machine Learning for Sequence-to-Function Approaches. In *Machine Learning and Big Data-enabled Biotechnology*, Wiley, pp. 103–129. Book chapter. doi.org/10.1002/9783527850532.ch4

Presentations

'26 **ISMB** (poster). Astra: A Mechanistic Multi-Task Learning Framework for Biochemically Consistent Enzyme Kinetic Parameter Prediction.

'26 **COBRA** (poster). BactoCat: A Hybrid Modeling Framework to Estimate In Vivo Enzyme Kinetics.

'25 **ICLR, Learning Meaningful Representations of Life Workshop** (poster). Bridging Sequence and Kinetics: Utilizing Multi-scale Representations for Genome-Scale Metabolic Models.

'24 **COBRA** (oral). kinGEMs: Advancing Enzyme-Constrained Genome-Scale Models with Deep Learning Predicted Kinetic Parameters.

’24 **COBRA** (poster). Dynamic Flux Balance Analysis Approach for Evaluating the Economic Potential of Bioproducts of Industrial Relevance.

’24 **AIChE Annual Meeting** (oral). Enhancing Compound-Protein Interaction Prediction with Confidence Assessment.

Teaching Experience

Teaching Assistant, University of Toronto

’25–’27 **CHE1147 and CHE1148: Chemical Data Science and AI for Applied Chemistry**, *graduate courses, Prof. Mohamad Moosavi and Prof. Benjamin Sanchez-Lengeling*
Designed and led tutorials on data science, machine learning, and deep learning for 60+ students per course. Supervised final projects that applied these methods to chemical engineering, chemistry, and biochemistry problems.

’25 **CHE222: Process Dynamics: Modeling, Analysis and Simulation**, *undergraduate course, Dr. Alexandre Tremblay*
Led numerical methods tutorials for 150+ students. Designed and supervised the final project on modeling chemical reactor dynamics.

Student Supervision

’26 **Hussain Alharbi**, *Summer Undergraduate Data Sciences Institute Research Program, UofT*

’25–’26 **Yuxi Zhang**, *Undergraduate Research Thesis (BCB430), UofT*

Honors and Awards

’24–Now **SECIHTI Scholarship**, Government of Mexico.

’24–Now **Connaught International Scholarship**, University of Toronto.

’24–Now **Junior Fellow**, Massey College, University of Toronto.

’23 **Mitacs Globalink Research Internship**. Funded research internship in Canada.

’20–’24 **Academic excellence**, ITESM. First in Class Honors. Top 5% GPA in the department every term. Academic Talent Scholarship.

Leadership and Service

’25–Now **Globalink Mentor**, *Mitacs*, Toronto
Guided 20+ international students through university onboarding and cultural adaptation during their summer internships.

’23 **Organizing Committee**, *BioMolecule Research Lab Symposium, ITESM*
Co-directed the inaugural edition with 100+ attendees. Led the scientific program and supervised event logistics.

’22 **Scientific Committee Lead**, *Latin Food 2022 Congress, AMECA*
Led the review of 500+ conference abstracts.

’21–’22 **Volunteer**, *Tutoring and non-profit work*, Puebla, Mexico
Tutored undergraduates in science and programming. Taught English to primary school students. Produced media for two non-profits (Indigenia Mundus, Al Diablo Las Tallas).

Skills

ML Deep learning, protein and genome language models, gradient-boosted trees, LLM agents, metabolic modeling optimization, constraint-based modeling

Code Python (PyTorch, TensorFlow, Hugging Face, scikit-learn, XGBoost, LangGraph, Weights & Biases), MATLAB, Git, Bash, Slurm, HPC, LaTeX

Languages Spanish (native), English (full professional, TOEFL C1), French (elementary, DELF A2 with merit)